

HASSAN NAEEM

AI/ML Engineer | Agentic AI & AWS-Native Systems

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PROFESSIONAL SUMMARY

AI/ML Engineer with 5+ years designing, building, and deploying production-grade agentic AI systems in Python on AWS for enterprise and startup clients. Hands-on across the full delivery lifecycle, discovery, architecture, and production, spanning agentic AI development, multi-agent orchestration, RAG pipelines, and API-driven services in data-centric environments. Deep expertise in Python and AWS-native development (Lambda, Bedrock, S3, IAM) with Terraform infrastructure automation, CI/CD, and cloud security best practices. Client-facing engineer who scopes ambiguous problems with stakeholders and ships real systems, not prototypes.

CORE SKILLS

Agentic AI & LLM: Agentic AI development, multi-agent orchestration, RAG pipelines, LLM API integration (Claude / AWS Bedrock, OpenAI, Vertex AI), LangChain, VAPI.ai voice agents, agentic workflow design, prompt engineering

Cloud & DevOps: AWS (Lambda, S3, EC2, Bedrock, CloudWatch, IAM), Terraform (IaC), Docker, Kubernetes, CI/CD automation, cloud architecture & security, GCP / Vertex AI, Grafana

Languages: Python, TypeScript, JavaScript (ES6+), SQL

Backend & APIs: FastAPI, Node.js, Django, REST API development, GraphQL, WebSockets, RabbitMQ

Databases: PostgreSQL, MongoDB, MySQL, Firebase Firestore, Redis

Frontend: React.js, Next.js, React Native, Angular, Redux, Tailwind CSS

Practices: Agile / Scrum delivery, code reviews, paired collaboration, production reliability

PROFESSIONAL EXPERIENCE

Software Engineer (Contract), Ally Financial

Aug 2026 – Present

Detroit Metropolitan Area (Hybrid) · AI Agents & LLM Automation for Banking

- Building AI agents and LLM-based automation systems for enterprise platforms, developing intelligent workflows that streamline internal engineering and operations processes.
- Developing backend services and agent orchestration using Python, FastAPI, and AWS, integrating LLM capabilities into production banking infrastructure with a focus on reliability and compliance.
- Collaborating with platform engineering teams on knowledge transfer, system design, and the rollout of AI-assisted tooling across the organization.

AI/ML Engineer, Independent / Freelance

Nov 2025 – Present

Remote · Agentic AI Systems & LLM Products

- Building a production agentic AI SaaS platform end-to-end, multi-agent discovery and outreach automation powered by the Claude API and RAG retrieval, on a Next.js / TypeScript / Supabase stack.
- Designing and deploying agentic AI systems on AWS (Bedrock / Claude) with Python FastAPI backends, multi-agent orchestration, RAG pipelines, and LLM voice agents (VAPI.ai).
- Advising on LLM and AWS cloud architecture, HIPAA-aware deployment, and production reliability and security for AI-driven products.

AI/ML Engineer (Contract), Peekly (AI Real Estate Platform)

Apr 2025 – Nov 2025

Remote · AI-Driven Property Recommendation & AR

- Architected an AI-driven property recommendation engine in Python, using Vertex AI and the Claude API to serve personalized recommendations over a high-throughput data pipeline (GCP + MongoDB).
- Built a hybrid React Native application (TypeScript, Node.js, Google Maps API) supporting immersive 3D and AR property exploration.
- Owned delivery from architecture through production deployment as lead engineer on the engagement.

AI/ML Engineer (Contract), Healthcare Automation Platform (Stealth Startup)**Nov 2024 – Apr 2025***Remote · HIPAA-Compliant Clinical Platform*

- Designed and deployed a HIPAA-compliant clinical platform end-to-end, building Python services that integrate Vertex AI and the Claude API into an automated prescription-suggestion engine and predictive resource-analytics service, reducing manual prescription review time.
- Built secure real-time patient–doctor communication (React, TypeScript, WebSockets, Firebase) on GCP with a MongoDB backend for scalable, compliant data management.
- Worked directly with clinical stakeholders to scope ambiguous requirements and translate them into production AI workflows.

AI/ML & Frontend Engineer, Culturefy Inc.**Mar 2023 – Oct 2024***Atlanta, GA (Remote) · AI-Powered HR Platform*

- Architected and deployed a production AI agent platform integrating VAPI.ai (LLM voice), HeyGen, AWS Transcribe, and OpenAI APIs (React.js, Next.js, Node.js), enabling voice-command interactions, autonomous content generation, and real-time dashboard control; scaled to 100K+ users.
- Designed agentic recruitment workflows with LLM-driven candidate ranking, interview-scheduling automation, and offer-letter generation; integrated the BambooHR API for real-time pipeline synchronization.
- Built a campaign-intelligence module in Python (GraphQL, StackAdapt) using ML-driven audience segmentation to optimize content distribution.
- Implemented CI/CD pipelines on AWS (EC2, S3, CloudWatch) with Docker and Kubernetes, reducing release-cycle time and improving reliability.

Software Engineer, Binate Digital**Jan 2022 – Feb 2023***Austin, TX · Digital & Mobile Platforms Agency*

- Led frontend development of the Take Charge EV-charging platform (React Native, TypeScript, Python/Django, Firebase, WebSockets) with real-time driver tracking, Authorize.net payments, and 100ms audio calling; shipped to production.
- Built the DateJar matchmaking app (React Native, GraphQL, Firebase, Agora SDK) with real-time messaging, audio/video calling, and in-app purchases across iOS and Android.
- Optimized the Nouvoprix marketplace (React Native, Django, GraphQL) with advanced product search and real-time messaging.

Associate Software Engineer, MiniBig Technologies**Sep 2020 – Dec 2021***Dubai, UAE · Early-Stage E-Commerce & Logistics Startup*

- Developed the GroceryDone4u e-commerce platform (React Native, Node.js, TypeScript, Firebase, Stripe, Google Maps) with multi-store inventory, food ordering, and real-time delivery tracking.
- Built the Pacsend peer-to-peer delivery platform (Angular, TypeScript, Node.js, Firebase, GraphQL, Stripe) with real-time tracking and payment processing.

EDUCATION

M.S., Computer Science, Concordia University Saint Paul